Chapter 2 — Unit & Combat Systems

This chapter defines every mechanic and system that constitutes a unit's identity, the structure of a combat turn, the resolution of combat itself, and the weapon system those combats draw from. It covers Layers 02 through 05. These four layers are the mechanical skeleton of the entire game: before any battle can be played, units must exist as data objects with complete stat profiles (Layer 02); before any unit can act, the game must know whose turn it is and what a single action costs (Layer 03); before any damage can be dealt, the formulas for hitting, avoiding, and calculating damage must be in place (Layer 04); and before any weapon can be swung, weapons themselves must exist as typed, tiered objects that interact with those formulas (Layer 05). Nothing in Chapters 3 through 7 can be built until this chapter is complete and stable.

Layer 02 — Unit Identity Core

Dependency: Layer 01 (Grid & Spatial Core) must be complete before this layer begins. Stats are meaningless without the grid they will act on, and the unit data model is the object every subsequent system reads from.

UC-01 — Class Architecture | Cat 1 | Unit & Class System

- [ ] Class data structure:
  - [ ] Each class is a named data record with the following fields, all required:
    - [ ] Class name (string, display name for UI)
    - [ ] Class type tag (one of: INFANTRY, CAVALRY, FLYING, ARMOURED, MAGE, HEALER, THIEF, MONSTER)
    - [ ] Movement range (integer, base tiles per turn)
    - [ ] Movement type tag (one of: FOOT, HORSE, FLYING, ARMOURED) — used by terrain passability rules

        - [ ] Weapon type whitelist (list of weapon type codes this class can equip; minimum one entry)
        - [ ] Stat growth modifier array (per-stat % modifiers applied on top of unit's base growth rates at level-up)
        - [ ] Stat cap array (maximum value for each stat in this class; 8 values: HP, Str, Mag, Skl, Spd, Def, Res, Con, Niyati)
        - [ ] Promotion eligibility flag (boolean: true if class can be promoted from)
        - [ ] Promoted class flag (boolean: true if this class is the result of a promotion)
        - [ ] Promotion target list (array of class IDs this class can promote into; empty if promotion not applicable)
        - [ ] Base class reference (for promoted classes only: the class ID this was promoted from; used for stat retention logic)
        - [ ] Class ability list (array of ability IDs passively active while unit is in this class; can be empty)
        - [ ] Map sprite ID (the sprite asset used for this class on the battle map)
        - [ ] Combat animation set ID (the animation asset set for attack, idle, death, etc.)
    - [ ] Class records are static — they do not change at runtime. All class data is loaded at game initialisation from a class definition table.
    - [ ] Minimum viable class set for Layer 02 (further classes may be added in content passes; these must exist for the game to function):
        - [ ] Lord (INFANTRY, FOOT, SWORD, promotable)
        - [ ] Cavalier (CAVALRY, HORSE, SWORD/LANCE, promotable)
        - [ ] Knight (ARMOURED, ARMOURED, LANCE/SWORD, promotable)
        - [ ] Myrmidon (INFANTRY, FOOT, SWORD, promotable)
        - [ ] Fighter (INFANTRY, FOOT, AXE, promotable)
        - [ ] Archer (INFANTRY, FOOT, BOW, promotable)
        - [ ] Mage (MAGE, FOOT, ANIMA_TOME, promotable)
        - [ ] Monk/Cleric (HEALER, FOOT, STAFF/LIGHT_TOME, promotable)

        - [ ] Thief (THIEF, FOOT, SWORD, promotable)
        - [ ] Pegasus Knight (FLYING, FLYING, LANCE, promotable)
        - [ ] Wyvern Rider (FLYING, FLYING, LANCE/AXE, promotable)
        - [ ] Shaman (MAGE, FOOT, DARK_TOME, promotable)
        - [ ] Soldier (INFANTRY, FOOT, LANCE, promotable — enemy-only base class or generic ally)
        - [ ] At least one promoted form for each promotable class above
- [ ] Unit-to-class binding:
    - [ ] Each unit record contains a current_class_id field pointing to exactly one class record.
    - [ ] When a unit's current_class_id changes (promotion), all stat caps are updated to the new class's cap array immediately.
    - [ ] When a unit's current_class_id changes, all weapon type whitelists are updated to the new class's weapon type whitelist immediately.
    - [ ] When a unit's current_class_id changes, all passive class abilities are removed and replaced with the new class's ability list.
    - [ ] Current HP does not change on promotion. If current HP exceeds the new class's HP cap (rare edge case), current HP is clamped to the new cap.
- [ ] Class type tag usage rules:
    - [ ] CAVALRY and FLYING types unlock the Canto ability (implemented in Layer 08); this is read from class type, not from the class ability list.
    - [ ] FLYING type grants terrain immunity (implemented in Layer 08); this is read from class type.
    - [ ] ARMOURED type confers no special rule in this layer; it is read by terrain passability logic (Layer 01) and by effective weapon logic (Layer 05).
    - [ ] MONSTER type is an enemy-exclusive tag. It is read by the Bishop Monster Slayer ability (Layer 08) and effective weapon logic (Layer 05).

SS-02 — Strength (Physical Damage) | Cat 1 | Stat System

- [ ] Stat definition:
  - [ ] Strength (Str) is an integer stat on every unit record. Minimum value: 0. Maximum value: class cap for Str.
  - [ ] Strength contributes to physical damage output per the damage formula (CC-06, Layer 04).
  - [ ] Strength does not contribute to magical damage. Magical damage uses the Magic stat (SS-03).
  - [ ] Strength stat cap is enforced at all times: any operation that would raise Str above the class cap for the current class silently clamps to the cap value.
- [ ] Stat growth:
  - [ ] Each unit has a str_growth_rate (integer percentage, 0–100). On every level-up, one random roll (0–99) is made. If the roll is below str_growth_rate, Str increases by 1. Otherwise, Str does not increase this level.
  - [ ] Growth rate is a fixed unit property. It does not change with class changes.
  - [ ] Growth rate modifiers from class (stat growth modifier array on the class record) are added to the base growth rate at the time of the level-up roll only. They are not permanently stored on the unit.

SS-03 — Magic (Magical Damage) | Cat 1 | Stat System

- [ ] Stat definition:
  - [ ] Magic (Mag) is an integer stat on every unit record. Minimum value: 0. Maximum value: class cap for Mag.
  - [ ] Magic contributes to magical damage output per the damage formula (CC-06, Layer 04).
  - [ ] Magic does not contribute to physical damage. Physical damage uses the Strength stat (SS-02).
  - [ ] Magic stat cap is enforced at all times: any operation that would raise Mag above the class cap silently clamps to the cap value.
  - [ ] For non-mage, non-healer classes, Magic stat cap is typically 0 or a very low value set in the class record. If the cap is 0, the Magic stat is functionally absent for that class.

The stat field still exists on the unit record; it simply cannot grow above 0.

- [ ] Stat growth:
  - [ ] Each unit has a mag_growth_rate (integer percentage, 0–100). Level-up roll logic is identical to SS-02.

SS-04 — Skill (Hit Rate + Crit Rate) | Cat 1 | Stat System

- [ ] Stat definition:
  - [ ] Skill (Skl) is an integer stat on every unit record. Minimum value: 0. Maximum value: class cap for Skl.
  - [ ] Skill contributes to Hit Rate and Critical Rate per the formulas in CC-04 and the crit formula respectively (Layer 04).
  - [ ] Skill stat cap is enforced at all times.
- [ ] Stat growth:
  - [ ] Each unit has a skl_growth_rate (integer percentage, 0–100). Level-up roll logic is identical to SS-02.

SS-05 — Speed (Double Attack + Avoid) | Cat 1 | Stat System

- [ ] Stat definition:
  - [ ] Speed (Spd) is an integer stat on every unit record. Minimum value: 0. Maximum value: class cap for Spd.
  - [ ] Speed contributes to Avoid (CC-05) and to the double attack threshold check (CC-08), both defined in Layer 04.
  - [ ] Speed stat cap is enforced at all times.
- [ ] Attack Speed (AS) — derived value, not a stored stat:
  - [ ] Attack Speed is calculated at the moment of combat initiation: AS = Spd − max(0, weapon_weight − unit_Con).
  - [ ] AS is not stored on the unit record. It is recalculated each time it is needed.
  - [ ] AS minimum is 0. It cannot go below 0 regardless of weapon weight or Con value.
  - [ ] If no weapon is equipped (e.g., during a staff use), AS equals Spd.
- [ ] Stat growth:

    - [ ] Each unit has a spd_growth_rate (integer percentage, 0–100). Level-up roll logic is identical to SS-02.

SS-07 — Defense (Physical Damage Reduction) | Cat 1 | Stat System

- [ ] Stat definition:
    - [ ] Defense (Def) is an integer stat on every unit record. Minimum value: 0. Maximum value: class cap for Def.
    - [ ] Defense reduces incoming physical damage per the damage formula (CC-06, Layer 04).
    - [ ] Defense does not reduce magical damage. Magical damage reduction uses Resistance (SS-08).
    - [ ] Defense stat cap is enforced at all times.
- [ ] Stat growth:
    - [ ] Each unit has a def_growth_rate (integer percentage, 0–100). Level-up roll logic is identical to SS-02.

SS-08 — Resistance (Magic Damage Reduction) | Cat 1 | Stat System

- [ ] Stat definition:
    - [ ] Resistance (Res) is an integer stat on every unit record. Minimum value: 0. Maximum value: class cap for Res.
    - [ ] Resistance reduces incoming magical damage per the damage formula (CC-06, Layer 04).
    - [ ] Resistance does not reduce physical damage. Physical damage reduction uses Defense (SS-07).
    - [ ] Resistance stat cap is enforced at all times.
- [ ] Stat growth:
    - [ ] Each unit has a res_growth_rate (integer percentage, 0–100). Level-up roll logic is identical to SS-02.

SS-10 — Movement (Mov) | Cat 1 | Stat System

- [ ] Stat definition:
    - [ ] Movement (Mov) is an integer stat on every unit record. Minimum value: 1. Maximum value: class cap for Mov.

    - [ ] Movement determines the maximum number of tiles a unit can traverse in a single move action, subject to terrain cost rules (MT-02, Layer 01).
    - [ ] Movement is read directly from the unit's current class record as the class's movement range field. It is not stored separately on the unit record unless a temporary modifier is active.
    - [ ] Movement does not grow through level-ups in the standard sense. It is class-determined. Promotion to a new class may result in a higher movement value if the promoted class has a higher movement range.
    - [ ] Temporary movement modifiers (from status effects, staves, or terrain) are applied as additive offsets to the base movement value. These offsets are stored on the unit record as a temp_mov_modifier field and are cleared when the effect expires.
    - [ ] Effective movement for a given turn = base class movement + temp_mov_modifier. Minimum effective movement is 1.

SS-01 — HP / Hit Points → Human HP Frame | Cat 2 | Stat System

DESIGN INTENT
This mechanic serves ES2 ("Feel like someone who finds people in the wreckage of war and carries them") and ES3 ("Feel like the conscience of a kingdom, carrying the weight of every life your war touches"). In FE GBA, HP is a tactical resource — a countdown to a unit loss. This version retains the identical mechanical function of HP (unit dies at 0, Lord at 0 = game over) but adds a framing layer that makes low HP feel like a person is in danger, not a resource is depleted. The modification is entirely in the presentation layer and reaction system — the HP formula and death threshold are unchanged.

- [ ] HP stat definition:
    - [ ] HP (Hit Points) is an integer stat on every unit record. Current HP minimum: 0. Current HP maximum: equal to the unit's max_HP value.
    - [ ] max_HP is the unit's maximum HP capacity. It is determined by the unit's base HP stat (set at

recruitment/spawn) and grows through level-ups.
    - [ ] When a unit's current HP reaches exactly 0, the unit is dead. Death handling is defined in Layer 07 (permadeath system, UM-01).
    - [ ] HP stat cap is enforced: current HP cannot exceed max_HP at any time. Any heal that would exceed max_HP is clamped to max_HP.
    - [ ] HP growth: Each unit has an hp_growth_rate (integer percentage, 0–100). On level-up, HP gains are multiplied relative to other stats — HP typically gains 2 points per successful growth roll rather than 1. The exact HP gain per successful roll is defined in the class record's hp_gain_on_levelup field (default: 2, range: 1–5 depending on class).
- [ ] HP threshold tracking:
    - [ ] The system continuously monitors each unit's current HP relative to their max_HP.
    - [ ] Define three HP threshold states per unit:
        - [ ] NORMAL: current HP > 50% of max_HP
        - [ ] INJURED: current HP is between 31% and 50% of max_HP (inclusive)
        - [ ] CRITICAL: current HP is ≤ 30% of max_HP
    - [ ] Threshold state is recalculated after every HP change (damage received, healing applied).
    - [ ] When a unit's state transitions from NORMAL to INJURED, no event fires beyond updating the portrait state.
    - [ ] When a unit's state transitions from INJURED to CRITICAL, or from NORMAL directly to CRITICAL (large single-hit damage), fire the hp_critical event for that unit.
- [ ] Portrait state machine (Human HP Frame visual):
    - [ ] Every named unit has a portrait asset for each of the three HP states: portrait_normal, portrait_injured, portrait_critical.
    - [ ] Generic/unnamed enemy units are not required to have distinct portrait states for INJURED. They must have a CRITICAL state portrait or fall back to a generic class silhouette.
    - [ ] The battle HUD unit info panel (built in Layer 20, UI-01) reads the unit's current HP threshold state and displays the

corresponding portrait asset.
    - [ ] Portrait state updates visually without delay: on the frame that HP change resolves and the new threshold state is computed, the portrait asset swaps immediately.
    - [ ] Portrait state does not affect any combat stat or formula. It is presentation only.
- [ ] Low HP reaction lines — character-specific dialogue:
    - [ ] Each named unit has a low_hp_lines[] array. This array contains a minimum of 3 and a maximum of 5 dialogue strings, authored per-character.
    - [ ] These lines are character-specific and not generic system strings. They are written in the character's narrative voice as defined in their character document.
    - [ ] Trigger rule: When a unit's HP threshold state transitions to CRITICAL for the first time in a battle, select and play a line from their low_hp_lines[] array.
        - [ ] The first transition to CRITICAL in a given battle uses the line at index 0 of the array (the "first time critical" line, which is always the most narratively resonant line for that character).
        - [ ] If the unit recovers HP above the CRITICAL threshold (healed), then drops back to CRITICAL again in the same battle, a subsequent line is selected from the remaining array entries (index 1 onward), cycling if needed. This prevents the first-time line from repeating.
    - [ ] Delivery: The line is displayed as a speech bubble or dialogue overlay anchored to the unit's portrait on the battle HUD. It is not a full-screen dialogue box. Duration: 2.5 seconds visible, then fades. The game does not pause for this line.
    - [ ] Unnamed/generic units: No low HP lines. No system fires for them.
- [ ] Adjacent unit reactions:
    - [ ] When the hp_critical event fires for a named unit, check all tiles within 4-directional range 1 (immediately adjacent — no diagonal) for allied units.
    - [ ] For each adjacent allied unit found, check the support relationship between that adjacent unit and the critical

unit.
        - [ ] If support rank is B or A: the adjacent unit plays a reaction line from their adjacent_reaction_lines[critical_unit_id][] array (authored per-pair at B and A rank respectively). These are specific, personal lines.
        - [ ] If support rank is C: the adjacent unit plays a generic-but-character-voiced reaction line from their adjacent_reaction_lines_c[] array (one-line, reflects the character's personality archetype but is not relationship-specific).
        - [ ] If support rank is none (no support relationship exists between the pair): no reaction fires.
    - [ ] Only one adjacent unit reaction fires per hp_critical event, even if multiple adjacent units qualify. Priority: highest support rank fires first. If a tie, select randomly among the tied units.
    - [ ] Delivery: Identical to low HP lines — speech bubble on the reacting unit's portrait, 2.5 seconds, no pause.
    - [ ] This system is independent of the low HP reaction line. Both can fire on the same event: the critical unit says their low HP line, and simultaneously (or offset by 0.5 seconds) an adjacent ally reacts.
- [ ] Healer highlight system:
    - [ ] A healer-class unit (class type tag: HEALER, or any unit with a staff equipped and healing staff ability) that has can_act = true AND has at least one allied unit in CRITICAL state within their staff's range receives a soft_pulse_highlight applied to their portrait on the battle HUD.
    - [ ] soft_pulse_highlight is a visual effect only: a slow, low-intensity pulse on the portrait border. It is not a flashing warning. It is not accompanied by a sound effect.
    - [ ] The highlight is cleared when: the healer acts (can_act becomes false), the critical unit is healed out of CRITICAL state, or the turn ends.
    - [ ] This is not a recommendation system. The game does not suggest the heal. The highlight communicates only that the healer has noticed the situation. The player decides.

- [ ] Post-survival flag:
  - [ ] At chapter end, for each named unit that is alive: if at any point during that chapter their current HP was ≤ 4 (single-digit HP), set survived_critical_HP_flag[unit_id] = true.
  - [ ] At the start of the next chapter (during pre-battle preparation or chapter intro), if survived_critical_HP_flag[unit_id] = true for a unit, their first pre-battle dialogue line is replaced with their survived_near_death_line (authored per-character, one line per character).
  - [ ] Clear survived_critical_HP_flag[unit_id] after the post-survival line has been delivered.
  - [ ] If the unit did not have their line delivered (e.g., they were not shown in pre-battle dialogue that chapter), clear the flag at the start of the following chapter anyway. The moment passes.

EDGE CASES

- A unit takes damage in a single hit that drops them from NORMAL directly to 0 HP (instant kill). The hp_critical event does not fire because death handling (UM-01) pre-empts it. No low HP line is delivered. Death animation plays instead.
- A unit at CRITICAL HP is healed to NORMAL in the same enemy phase as the original hit (rare — would require an AI healer acting after the damage). The portrait reverts to NORMAL state. The low HP line has already been delivered; it is not retracted.
- A unit's max_HP is very low (e.g., 12 HP total). The 30% CRITICAL threshold is 3.6, which rounds to 4. All thresholds are computed using integer arithmetic with floor rounding.
- Adjacent unit reaction fires for an ally who has also just been damaged and is themselves in CRITICAL state. The reaction still fires. A unit in CRITICAL state can still call out to another.

SS-06 — Luck Stat → Niyati (Karma) Stat | Cat 2 | Stat System

DESIGN INTENT
This mechanic serves ES3 ("Feel like the conscience of a kingdom") and ES4 ("Feel like a burdened strategist forced to choose between clean principles and necessary evils"). In FE GBA, Luck is a silent statistical buffer — it marginally improves Hit Rate and reduces enemy Critical Hit chance. Players almost never think about it consciously. This version retains those exact mechanical functions but replaces the name, display, and the invisible nature of the stat with a visible, narratively legible representation tied to a unit's relationship with their own dharmic alignment. The stat does not function as a morality score — it does not reward "good" choices. It tracks whether a character is living in accordance with their declared nature and sworn commitments.

- [ ] Stat definition (mechanical function — unchanged from FE GBA Luck):
  - [ ] Niyati (Niy) is an integer stat on every unit record. Minimum value: 0. Maximum value: class cap for Niyati (typically 30, consistent with FE GBA Luck caps).
  - [ ] Niyati contributes to the Hit Rate formula (CC-04, Layer 04) in the same way FE GBA Luck does: Hit Rate includes a +Niyati term.
  - [ ] Niyati contributes to Critical Evasion: Critical Evasion = Niyati. This is subtracted from the attacker's Critical Hit Rate when computing whether a crit triggers.
  - [ ] Niyati stat cap is enforced at all times.
- [ ] Stat growth:
  - [ ] Each unit has a niyati_growth_rate (integer percentage, 0–100). Level-up roll logic is identical to SS-02.
  - [ ] Niyati growth through levels is the base growth path for this stat.
- [ ] Narrative modification events (Niyati delta system):
  - [ ] In addition to level-up growth, Niyati can shift up or down through specific named story events. These are stored in a niyati_event_table: a mapping of story_flag_id → {unit_id, delta_value}.

    - [ ] Maximum number of niyati modification events per character per playthrough: 4. This limit is enforced by design (content constraint, not a runtime check needed).
    - [ ] Delta values are small: +1 or −1 per event. No single event changes Niyati by more than 1.
    - [ ] Niyati modification events fire when the corresponding story_flag is set by the campaign event system (Layer 15). The delta is applied to the unit's Niyati value at that moment.
    - [ ] Niyati modifications from story events are capped by the same stat cap and stat minimum rules: Niyati cannot go below 0 or above class cap through story events.
    - [ ] Example event types that fire niyati modifications (exact event list is a content decision, not a runtime rule):
        - [ ] A character keeping a sworn promise: +1 Niyati for that character.
        - [ ] A character breaking a sworn promise: −1 Niyati for that character.
        - [ ] A character surviving something that should have killed them (probabilistic near-death, tracked by the survived_critical_HP_flag system from SS-01): +1 Niyati.
        - [ ] A character using a weapon designated as forbidden/adharmic in their backstory: −1 Niyati.
    - [ ] The game never explicitly tells the player that a story event changed a unit's Niyati. The player discovers it by reading the stat screen before and after, or by noticing the symbol state change (see display below).
- [ ] Niyati display:
    - [ ] The stat is labelled "Niyati" on the unit stat screen (not "Luck").
    - [ ] The numerical value is displayed (integer).
    - [ ] A niyati_symbol_state is computed from the ratio of current Niyati to the character's baseline Niyati (their Niyati value at the time of recruitment/creation):
        - [ ] current_niyati / base_niyati > 1.2 → symbol state: LOTUS_FULL
        - [ ] current_niyati / base_niyati between 0.8 and 1.2 (inclusive) → symbol state: LOTUS_HALF

        - [ ] current_niyati / base_niyati < 0.8 → symbol state: LOTUS_WITHERED
    - [ ] The appropriate symbol asset is displayed adjacent to the Niyati value on the stat screen. The symbol swaps when the state changes.
    - [ ] No tooltip explains the symbol. No in-game text describes what LOTUS_FULL or LOTUS_WITHERED means. The symbol is discoverable, not explained.
- [ ] Low Niyati visual effect:
    - [ ] If a unit's current Niyati is ≤ 2, apply shadow_portrait_overlay to their portrait in the battle HUD unit info panel (UI-01, Layer 20).
    - [ ] shadow_portrait_overlay is a subtle darkening shader over the portrait. It is not dramatic — a slight desaturation and shadow, not a complete overwrite.
    - [ ] The shadow_portrait_overlay is suppressed during any turn where the unit performs a dharmic_action_flag action in that turn. A dharmic_action_flag is set on actions including: healing an ally (staff use), performing a Rescue on another unit, and any action tagged as a promise-fulfillment narrative event by the campaign event system.
    - [ ] If a dharmic_action_flag action is performed, the shadow lifts for that turn and is restored at the start of the following turn if Niyati is still ≤ 2.
    - [ ] This effect has no tooltip, no UI explanation. It is purely visual.
- [ ] Niyati and the post-death honesty system (interaction with CC-10, Layer 04):
    - [ ] The True Hit system (CC-10) checks if the killing blow had True Hit < 30 to set unlikely_death = true. If the dead unit's Niyati was at LOTUS_WITHERED state at the time of death, the post-battle acknowledgment line (from CC-10) is modified: instead of the default scribe line, an additional line is added: "Their niyati had grown thin. The dice fell as the cosmos decided." This is an authored content line, not a generated string.

EDGE CASES

- A unit's base_niyati is 0 at recruitment (extremely rare). The niyati_symbol_state computation would divide by zero. If base_niyati = 0, default symbol state to LOTUS_HALF regardless of current value.
- A unit's Niyati grows through level-ups and pushes current_niyati well above base_niyati (e.g., base was 5, now 15 from growth). Symbol shows LOTUS_FULL. No events have fired. This is correct — the character has grown in alignment with their fate through the campaign, even without specific story events.
- Two story events fire in the same chapter: one +1 and one −1 for the same character. Both are applied. Net change: 0. The symbol state may not visually change. Both events are still internally logged.

SS-09 — Constitution (Con) → Human Limitation Frame | Cat 2 | Stat System

DESIGN INTENT
This mechanic serves ES3 ("Feel like the conscience of a kingdom, carrying the weight of every life your war touches"). The "carrying" is literal: Con determines who can physically carry (Rescue) whom. In FE GBA, failing to Rescue because of a Con mismatch produces a silent block with no acknowledgment — the action simply cannot be selected. This version retains the identical mechanical rules of Con (attack speed penalty, rescue threshold) but adds a voice layer: when a unit cannot lift another because they are physically unable to do so, the game says so in that character's own words. Physical limitation must be felt as human, not numerical.

- [ ] Stat definition:
  - [ ] Constitution (Con) is an integer stat on every unit record. Minimum value: 1. Maximum value: class cap for Con.
  - [ ] Constitution does NOT grow through level-ups in the standard probabilistic way. Con is a near-fixed stat per unit. Each unit has a base Con value set at recruitment that does not change except through promotion (some promoted classes adjust Con) or through stat booster items (ER-04, Layer 06, if implemented).

    - [ ] Con growth rate is 0 for virtually all units. Some promoted class transitions include a flat Con bonus applied once at promotion.
- [ ] Attack Speed penalty (Con interaction with weapon weight):
    - [ ] When a weapon is equipped, if weapon_weight > unit_Con, the unit suffers an Attack Speed penalty.
    - [ ] AS penalty = weapon_weight − unit_Con.
    - [ ] Effective AS = Spd − AS_penalty. Minimum effective AS is 0.
    - [ ] This calculation occurs at the time of combat initiation, not on equip. AS is not pre-computed and stored except as needed by the combat forecast UI.
- [ ] Rescue eligibility threshold:
    - [ ] A unit (the rescuer) can carry another unit (the rescued) only if: rescuer_Con ≥ rescued_Con − 10.
        - [ ] Equivalently: rescuer can carry rescued if the Con difference (rescued_Con − rescuer_Con) is 10 or less.
        - [ ] If rescued_Con − rescuer_Con > 10, the Rescue action is not available to this pair.
    - [ ] This threshold is checked when the player selects the Rescue action from the action menu. If the condition is not met, the target unit is not displayed as a valid Rescue target.
- [ ] Rescue attempt acknowledgment (Human Limitation Frame):
    - [ ] When a player attempts to initiate a Rescue action and no valid targets exist within range because all nearby units exceed the Con threshold, the game does not silently grey out the action.
    - [ ] Instead: the Rescue action remains selectable. When the player confirms the attempt and the system detects no valid targets due to Con mismatch (i.e., there are nearby allied units but none qualify due to Con), a one-line character-voiced response plays from the unit attempting the Rescue.
        - [ ] This response is drawn from the unit's rescue_fail_line[] array (1–2 authored lines per character). These lines reference the specific impossibility in the character's voice — they are not generic system messages.
        - [ ] Example line (character-specific): "I can't lift them. My body won't allow it."

        - [ ] Example line (for a small/light character): "They're too heavy. I'm sorry — I'm not strong enough for this."
    - [ ] Delivery: Speech bubble on the attempting unit's portrait. 2.5 seconds. No game pause.
    - [ ] This response only fires when: there is at least one adjacent allied unit that would be a valid Rescue candidate (correct adjacency, correct faction) but all fail the Con threshold check. If the action is greyed out because no allies are adjacent at all, no line fires.
- [ ] Heavy weapon visual acknowledgment:
    - [ ] Each weapon record has a weight_class tag: LIGHT, MEDIUM, or HEAVY. This is set at content creation time based on the weapon's weight value (thresholds: LIGHT = weight 1–4, MEDIUM = weight 5–9, HEAVY = weight 10+; exact thresholds to be confirmed by balance pass).
    - [ ] When a unit equips a weapon with weight_class = HEAVY, a modified idle animation loop plays for that unit on the map during the player phase.
    - [ ] The modified idle animation is the heavy_stance_idle variant: a subtle weight-shifting, slightly different posture from the standard idle. Duration loop identical to standard idle; only the animation frames differ.
    - [ ] This is a visual cue, not a stat tooltip. No text appears on screen. Players who notice the animation difference will feel the weight physically. Players who don't notice still experience the mechanical AS penalty.
    - [ ] If the weapon is unequipped or swapped for a lighter weapon, the idle animation returns to standard.
- [ ] Carried unit experience (the "being carried" state):
    - [ ] When a unit is in the rescued state (being carried by another unit — see Rescue mechanic, Layer 07), their battle HUD portrait shows a carried_portrait_state visual indicator (a small visual element on the portrait frame indicating they are being carried, not a stat readout).
    - [ ] If the carrying unit takes damage while carrying (i.e., an enemy attacks the carrier), a reactive line fires from the carried unit:
        - [ ] Source: the carried unit's set_down_line[] array (1 line per unit, authored individually).

    - [ ] This line fires at most once per unique carrier-carried pair per entire playthrough. A flag tracks this: set_down_line_fired[carrier_unit_id][carried_unit_id]. Once set, the line never plays again for that pair.
    - [ ] Delivery: Speech bubble on the carried unit's portrait, 2.5 seconds, no pause.

EDGE CASES

- A unit with Con 1 (minimum) attempts to Rescue a unit with Con 5. Threshold check: 5 − 1 = 4, which is ≤ 10. The Rescue is allowed. Con 1 can rescue Con 11 (11 − 1 = 10, exactly at threshold). Con 1 cannot rescue Con 12 (12 − 1 = 11, exceeds threshold).
- Both units in the pair have Con 1. The threshold check is 1 − 1 = 0, which is ≤ 10. Any unit can rescue any unit of equal or near-equal Con.
- A unit's Con changes via promotion (new class grants a flat Con bonus). Recheck all active rescue relationships involving this unit if any Rescue states are currently in effect. If the carrier's new Con now fails the threshold for a unit they're currently carrying, this is a contradiction — it cannot occur mid-carry. The Con change from promotion takes effect only at chapter start or in the pre-battle preparation phase, not mid-battle.
- The set_down_line fires for a carrier-carried pair, the pair is later separated, and in a different chapter the same carrier carries the same unit again. The flag set_down_line_fired[carrier][carried] is already set. No line fires. The moment is singular.

Layer 03 — Phase Structure & Turn Engine

Dependency: Layer 02 (Unit Identity Core) must be complete before this layer begins. The turn loop requires units to exist as data objects before they can act within it.

CC-01 — Player/Enemy/Allied Phase Structure → Named Commander Personality System (Sarathi) | Cat 2 | Core Combat

DESIGN INTENT
This mechanic serves ES5 ("Feel like the hidden hand behind the throne, quietly steering hot-headed leaders toward a future only you can see") and ES7 ("Feel like a calm center in chaos, guiding flawed heroes through a war they barely understand"). In FE GBA, the three-phase turn structure (Player Phase → Enemy Phase → Allied Phase) is straightforward. Allied NPC units act on a simple AI: attack adjacent enemies, prioritise low-HP or high-value targets. Their behaviour reads as competent but characterless. This version retains the three-phase structure exactly but replaces the Allied Phase AI with a personality-driven behaviour system (the Sarathi System) where each named allied commander acts according to a defined personality archetype. The player cannot command allied NPCs directly. Their job is to build a plan that accounts for each ally's nature.

- [ ] Phase structure — three-phase turn cycle:
  - [ ] A complete game turn consists of three phases executed in this fixed order: Player Phase → Enemy Phase → Allied Phase.
  - [ ] Each phase has a defined start event and a defined end event.
  - [ ] The Player Phase begins with a phase_start animation/banner display reading "Player Phase" (or localised equivalent). All player-controlled units have their can_act flag reset to true and their has_moved flag reset to false.
  - [ ] The Enemy Phase begins with a phase_start animation/banner display reading "Enemy Phase". All enemy-controlled units have their can_act flag reset to true and their has_moved flag reset to false. Enemy AI executes automatically (AI-01, Layer 10 — placeholder in this layer; the phase slot exists but AI execution is stubbed).
  - [ ] The Allied Phase begins with a phase_start animation/banner display reading "Allied Phase". All allied NPC units have their can_act flag reset to true and their

has_moved flag reset to false. Allied NPC AI executes according to the Sarathi System (see below).
    - [ ] Allied Phase only exists if at least one allied NPC unit is present on the map. If no allied NPCs are present, the turn cycle is two-phase only: Player Phase → Enemy Phase.
    - [ ] Turn counter increments by 1 at the end of the Allied Phase (or Enemy Phase if no Allied Phase). Turn counter starts at 1 on map load.
    - [ ] End condition for Player Phase: all player-controlled units have can_act = false (all have taken an action or Waited), OR the player explicitly selects "End Turn" from the pause menu. Either condition closes the Player Phase and begins the Enemy Phase.
    - [ ] End condition for Enemy Phase: all enemy units with can_act = true have completed their AI turn execution. The Enemy Phase end triggers the Allied Phase start (if applicable) or the next turn's Player Phase start.
    - [ ] End condition for Allied Phase: all allied NPC units with can_act = true have completed their AI turn execution. The Allied Phase end triggers the next turn's Player Phase start.
- [ ] Unit ownership and phase participation:
    - [ ] Every unit on the map has a faction tag: PLAYER, ENEMY, or ALLIED.
    - [ ] A unit participates in the phase corresponding to their faction tag.
    - [ ] Faction tag is set at unit spawn and cannot change mid-battle except through recruitment mechanics (Layer 16), which convert an ENEMY unit to PLAYER faction.
    - [ ] Units with can_act = false at the start of their faction's phase skip their turn silently. They do not perform any action.
- [ ] Named Commander Personality System — Sarathi:
    - [ ] This sub-system governs the behaviour of named allied NPC units during the Allied Phase. It does not apply to unnamed/generic allied units — those use a simple default allied AI (attack adjacent enemies).
    - [ ] A unit qualifies as a Named Commander if they have: a name, a portrait, a personality_archetype field set in their

unit record, and an emotional_state field. Units without these fields use the default allied AI.

- [ ] Personality archetypes — exactly one archetype is assigned to each Named Commander. The archetype is a fixed property of the unit and does not change across the campaign (though emotional_state severity can shift — see below):
  - [ ] VIRA (Heroic/Reckless): Prioritises the highest-threat enemy on the map regardless of positioning. Will advance into suboptimal terrain to reach the target. Will not retreat even at low HP. Never passes turn voluntarily.
  - [ ] DHIRA (Steady/Cautious): Only engages enemies from a position where terrain gives a defence or avoid bonus. Will not advance into open terrain with no cover. Will pass turn if no valid engagement position is reachable this turn. Will prioritise defensive repositioning over offensive engagement when both are possible.
  - [ ] KARUNA (Compassionate/Protective): Prioritises moving toward the nearest player-controlled unit at low HP (INJURED or CRITICAL state from SS-01) over attacking enemies. If no player unit is at low HP, attacks the nearest enemy. Will sacrifice own advantageous position to be adjacent to a damaged ally.
  - [ ] KRODHA (Wrathful/Vengeful): If a named player unit or named allied unit died in a previous turn of this battle, KRODHA fixates on the enemy unit responsible (the unit that dealt the killing blow). Ignores all other tactical priorities — moves toward that specific unit and attacks them regardless of positioning or threat level. If the target is unreachable this turn, moves as far as possible toward them. If the avenging target was already defeated, KRODHA reverts to a standard engagement priority.
  - [ ] LOBHA (Ambitious/Strategic): Prioritises actions that advance their own unit's position or experience. Will not support a player unit at the cost of a better

offensive opportunity for themselves. If two enemy targets are reachable, always targets the one with higher EXP yield. Will not Rescue player units under any circumstance.

- [ ] Emotional state machine per Named Commander:
  - [ ] Each Named Commander has an emotional_state variable with two components: a base state (defaults to NORMAL at the start of each chapter) and a severity level (NORMAL → ELEVATED → CRITICAL).
  - [ ] severity = NORMAL: standard archetype behaviour as defined above.
  - [ ] severity = ELEVATED: archetype behaviour is more pronounced. A VIRA at ELEVATED charges more aggressively — will move toward a target even if it triggers a counterattack from a second enemy en route. A DHIRA at ELEVATED refuses to engage unless terrain bonus is at least +20 Avoid.
  - [ ] severity = CRITICAL: archetype behaviour becomes extreme and may diverge from standard tactical sense entirely. A VIRA at CRITICAL attacks the strongest enemy on the map, regardless of HP, regardless of combat forecast. A KRODHA at CRITICAL ignores the objective entirely and refuses any action that is not attacking the unit they are focused on.
  - [ ] severity escalation triggers:
    - [ ] A named player unit dies in the same battle → all allied commanders receive severity + 1 escalation.
    - [ ] The commander is reduced to CRITICAL HP (SS-01 threshold) → severity + 1 escalation for that commander.
    - [ ] A successful Vakya (counsel action, Layer 16, Mechanic E-01) is performed on the commander → severity resets toward NORMAL by 1 step.
  - [ ] severity is capped at CRITICAL. It cannot escalate beyond CRITICAL.
  - [ ] personality_drift_flag: If a VIRA commander has experienced at least 2 named unit deaths in the current campaign (tracked in the death registry,

Cluster A), their archetype behaviour includes a KRODHA component — after their standard VIRA attack, if the killing blow target was responsible for a named death, the KRODHA fixation activates as a secondary priority. This flag is set once per character per campaign and does not reset.

- [ ] Behaviour priority tree — executed each turn for each Named Commander during Allied Phase:
  - [ ] Step 1: Is a severity = CRITICAL state active? → Execute extreme archetype behaviour. Ignore all other steps.
  - [ ] Step 2: Is a guidance_flag set from a Strategic Counsel action this chapter (see Layer 16)? → Modify behaviour per the guidance instruction. Continue to step 3 if the guidance does not fully define action.
  - [ ] Step 3: Is the optimal position for this archetype reachable this turn? → Move to optimal position and engage from there.
  - [ ] Step 4: Default archetype behaviour → attack highest-priority target per archetype logic, moving as far toward them as movement allows.
  - [ ] Step 5: If no valid attack is possible and archetype does not mandate passing → Wait (end turn in place).
- [ ] Anticipation UI — pre-Allied Phase portrait line:
  - [ ] Before the Allied Phase banner animation completes, a brief portrait-and-line display appears for each Named Commander who will act this turn.
  - [ ] Display order: portrait appears at bottom of screen in sequence, each for 1.5 seconds, overlapping the phase transition animation.
  - [ ] Each Named Commander displays: their portrait + a single authored anticipation line reflecting their current emotional_state and archetype.
  - [ ] Anticipation lines are authored per-archetype per-severity (minimum 3 lines per combination). The line selected is based on current archetype + severity.
  - [ ] Examples of tone (not exact lines):
    - [ ] VIRA, NORMAL: A line expressing eagerness for the fight.

        - [ ] DHIRA, NORMAL: A line expressing calculation or patience.
        - [ ] KRODHA, CRITICAL: A line focused entirely on the unit they are fixated on.
    - [ ] This display is not a tutorial or tooltip. It is a character expression moment. Players learn to anticipate ally behaviour by reading these lines over multiple chapters.
    - [ ] The anticipation display is skippable — pressing confirm/advance dismisses it immediately and the Allied Phase proceeds.
  - [ ] Failure moments — when personality causes a bad outcome:
    - [ ] When a Named Commander's personality-driven action leads to a negative outcome (e.g., VIRA charges into a trap, takes severe damage; KRODHA abandons a map objective to pursue revenge), the game does not interrupt or apologise.
    - [ ] If another Named Commander is within 3 tiles of the commander who made the bad decision, and they have a C-rank or higher support relationship, that other commander plays a single reaction line acknowledging the situation (1 line, authored per-pair, plays once per chapter per pair for this event type).
    - [ ] The player must respond to the consequences. The system does not offer a rewind or a warning.
  - [ ] Named Commander permadeath rule:
    - [ ] Named allied commanders follow permadeath rules identical to player units (Layer 07, UM-01). If a Named Commander reaches 0 HP, they are dead.
    - [ ] Named Commander death fires all Death Ripple events (Cluster A, UM-01 dependencies) through the same system as player unit deaths.
    - [ ] After a Named Commander dies, all remaining Named Commanders who had a support relationship with the dead commander receive a severity escalation relevant to their archetype at the start of the next Allied Phase (KRODHA archetypes escalate

sharply; DHIRA may become more cautious; specific escalation per-archetype is a content design decision).

- [ ] Unnamed/generic allied units:
  - [ ] Allied units without a personality_archetype field use the simple default allied AI: move toward nearest enemy, attack if in range, prioritise the target with lowest HP when multiple are reachable, Wait if no enemy is reachable.
  - [ ] Generic allied units do not participate in the anticipation UI.
  - [ ] Generic allied units do not fire Death Ripple events on death — their deaths are not narratively tracked.

EDGE CASES

- All Named Commanders on the Allied Phase have severity = CRITICAL and are engaged in fixated behaviours that conflict with each other (e.g., two KRODHA commanders are targeting different units). Each resolves their behaviour independently. There is no conflict resolution between commanders. The player must manage the resulting chaos.
- A Named Commander's personality-driven movement would place them on the same tile as a player unit. Standard movement blocking rules apply — units cannot share tiles. The commander pathfinds to the nearest reachable tile adjacent to their intended target.
- The Allied Phase has no allied units because all Named Commanders died in previous turns. The Allied Phase is skipped. Turn cycle becomes two-phase.
- A Named Commander has KRODHA fixation but the target unit they are fixated on has already been defeated (by a player unit, for example). See the KRODHA definition — revert to standard engagement priority. No special event fires.

CC-02 — Unit Action Economy (One Action Per Unit) | Cat 1 | Core Combat

- [ ] Action economy definition:
  - [ ] Each unit on the map has exactly one action available per phase. An "action" is defined as any of the following

mutually exclusive activities performed after movement: Attack, Use Item, Staff Use, Trade, Rescue, Drop, or Wait.
    - [ ] Movement is not an action. A unit may move and then take an action. A unit may also take an action without moving (by selecting an action from their starting tile). A unit may not take an action and then move (movement must precede action, not follow it, with the exception of Canto — defined in Layer 08).
    - [ ] "Wait" is an explicit action. Selecting Wait ends the unit's turn without performing any other action. A unit that has moved but not yet acted may select Wait to end their turn in their current position.
    - [ ] Per-turn state tracking per unit:
        - [ ] has_moved (boolean): set to true when the unit completes a move. Reset to false at the start of their faction's phase.
        - [ ] can_act (boolean): set to false when the unit completes an action (including Wait). Reset to true at the start of their faction's phase.
        - [ ] Once can_act = false for a unit, they are "done" for this turn. Their map sprite displays as greyed-out/desaturated to visually communicate that they have acted.
    - [ ] A unit cannot be selected for a second action in the same phase once can_act = false.
    - [ ] Movement confirmation and cancellation:
        - [ ] While a unit is in the "moved but not yet acted" state (has_moved = true, can_act = true), the player may cancel the move and return the unit to their original position — provided the player has not yet selected an action.
        - [ ] Once an action is selected and confirmed (attack committed, item used, Wait selected), the move cannot be cancelled.
        - [ ] The cancellation window ends the moment an action type is chosen from the action menu. Navigation within the action menu (before selecting) does not commit the action.
- [ ] Greyed-out visual:

    - [ ] When a unit's can_act = false, their map sprite applies a desaturation/greyscale overlay. The overlay is removed at the start of their next phase when can_act resets.
    - [ ] The greyed-out state is applied to the sprite layer that renders on the map, not to any other UI element (portrait, stat panel, etc.).
- [ ] Player phase end trigger (interaction with phase structure):
    - [ ] The Player Phase ends when either: all player-faction units have can_act = false, OR the player confirms "End Turn" from the pause menu.
    - [ ] If the player selects "End Turn" while some player units still have can_act = true, those units are treated as if they Waited. Their can_act becomes false before the Enemy Phase begins.

SYS-04 — Pause / Menu Navigation System | Cat 4 | Input & Core Engine

- [ ] Pause menu accessibility:
    - [ ] At any point during the Player Phase (not during an animation, not mid-combat resolution, not during a dialogue scene), the player can open the Pause Menu via a dedicated input.
    - [ ] The Pause Menu is not accessible during Enemy Phase execution or Allied Phase execution.
    - [ ] The Pause Menu is not accessible while a unit is in the "moved but not yet acted" state (this state uses the Action Menu — see UI-05, Layer 20). These are distinct menus.
    - [ ] Pause Menu is not accessible while any animation, cutscene, dialogue sequence, or battle sequence is playing.
- [ ] Pause Menu contents:
    - [ ] The Pause Menu contains exactly these top-level options, in this order:
        - [ ] End Turn: Closes the pause menu and ends the Player Phase (see CC-02 for End Turn handling).
        - [ ] Unit List: Opens a read-only list of all player-controlled units and their current stats. Selecting a unit from the list centres the camera on them on the map. Does not select the unit for action.

        - [ ] Map/Objectives: Displays the current map objective(s) and win/loss conditions for the current chapter.
        - [ ] Options: Opens the Settings / Options Menu (defined fully in Layer 20, UI-10). Settings made here persist immediately.
        - [ ] Suspend Save: Triggers a suspend save of the current game state (Layer 19, SAV-01). After the save completes, the game returns to the map exactly as it was.
        - [ ] Quit to Main Menu: After a confirmation prompt ("Are you sure? Unsaved progress will be lost."), returns to the main menu without saving the battle state.
    - [ ] "Quit to Main Menu" requires a two-step confirmation: select → confirmation dialog with Yes/No → if Yes, quit. If No, return to Pause Menu.
    - [ ] "End Turn" does not require a confirmation prompt. It executes immediately.
- [ ] Pause menu navigation:
    - [ ] Navigation within the Pause Menu is directional-input only (D-pad / analog stick). No pointer/mouse required (mouse/KB support is an accessibility concern handled in Layer 22).
    - [ ] The Pause Menu can be closed (returning to the battle map) via the designated Cancel/Back input at any time, returning to the state that existed before the menu was opened.
    - [ ] If the Pause Menu is opened from a sub-state (e.g., player is hovering over a unit that hasn't acted yet), closing the Pause Menu returns to that sub-state exactly — cursor position is preserved.
- [ ] Action Menu distinction:
    - [ ] The Action Menu (the menu that appears when a unit has moved and is selecting their action) is NOT the Pause Menu. It is a separate, context-specific menu defined in UI-05 (Layer 20). The Pause input does not open the Action Menu. The Action Menu's cancel/back input closes the Action Menu, not the Pause Menu.

Layer 04 — Core Combat Engine

Dependency: Layer 03 (Phase Structure & Turn Engine) must be complete before this layer begins. All combat formulas require units to exist (Layer 02) and require a turn context to execute within (Layer 03).

CC-04 — Hit Rate Formula | Cat 1 | Core Combat

- [ ] Hit Rate calculation:
  - [ ] Hit Rate is a per-combat-pair derived value computed fresh at the start of each combat exchange. It is not stored persistently.
  - [ ] Formula (attacker perspective):
    - [ ] Attacker Hit = (Skill × 2) + Niyati + Weapon Hit Bonus + Weapon Triangle Hit Bonus
    - [ ] Where:
      - [ ] Skill = attacker's Skl stat (SS-04)
      - [ ] Niyati = attacker's Niyati stat (SS-06)
      - [ ] Weapon Hit Bonus = the Hit stat on the equipped weapon (from weapon record)
      - [ ] Weapon Triangle Hit Bonus = +15 if attacker has weapon triangle advantage over defender's weapon type; −15 if disadvantage; 0 if neutral or no triangle applies (WS-01 / WS-02 in Layer 05)
  - [ ] Attacker Hit is an integer. No fractional values.
  - [ ] Attacker Hit minimum is 0. It cannot be negative regardless of modifiers.
  - [ ] Attacker Hit has no maximum cap. It can theoretically exceed 100.
  - [ ] Attacker Hit feeds into the True Hit calculation (CC-10).

CC-05 — Avoid Formula | Cat 1 | Core Combat

- [ ] Avoid calculation:
  - [ ] Avoid is a per-unit derived value computed fresh at the start of each combat exchange. It is not stored persistently.
  - [ ] Formula (defender perspective):

        - [ ] Avoid = (Attack Speed × 2) + Niyati + Terrain Avoid Bonus
        - [ ] Where:
            - [ ] Attack Speed = defender's effective AS at the time of this combat (Spd − max(0, weapon_weight − Con), as defined in SS-05)
            - [ ] Niyati = defender's Niyati stat (SS-06)
            - [ ] Terrain Avoid Bonus = the Avoid bonus granted by the terrain tile the defending unit currently occupies (MT-02, Layer 01)
    - [ ] Avoid is an integer. No fractional values.
    - [ ] Avoid minimum is 0. It cannot be negative.
- [ ] Displayed Hit Rate (combat forecast):
    - [ ] The displayed hit percentage shown on the combat forecast UI (UI-03, Layer 20) is the True Hit value, not the raw Attacker Hit − Avoid difference.
    - [ ] True Hit computation is defined in CC-10.
    - [ ] Raw Attacker Hit − Avoid is an internal intermediate value. It is never displayed to the player directly.
    - [ ] Display hit % = max(0, min(100, True_Hit_value)) — clamped to 0–100 for display. The internal True Hit value can exceed 100 or be negative; these are clamped at display time only, not at RNG evaluation time.

CC-06 — Damage Calculation | Cat 1 | Core Combat

- [ ] Damage formula — physical attack:
    - [ ] Physical Damage = max(0, Attacker_Strength + Weapon_Might + Weapon_Triangle_Damage_Bonus − Defender_Defense − Terrain_Defense_Bonus)
    - [ ] Where:
        - [ ] Attacker_Strength = attacker's Str stat (SS-02)
        - [ ] Weapon_Might = the Might (Mt) value on the equipped weapon
        - [ ] Weapon_Triangle_Damage_Bonus = +1 if attacker has weapon triangle advantage; −1 if disadvantage; 0 if neutral (WS-01 / WS-02, Layer 05)
        - [ ] Defender_Defense = defender's Def stat (SS-07)

        - [ ] Terrain_Defense_Bonus = the Defense bonus granted by the defender's current terrain tile (MT-02, Layer 01)
    - [ ] Physical damage minimum is 0. A unit can never be healed by taking a physical attack.
    - [ ] Physical damage applies when the attacker is using a weapon tagged as physical (weapon_damage_type = PHYSICAL in the weapon record).
- [ ] Damage formula — magical attack:
    - [ ] Magical Damage = max(0, Attacker_Magic + Weapon_Might + Magic_Triangle_Damage_Bonus − Defender_Resistance)
    - [ ] Where:
        - [ ] Attacker_Magic = attacker's Mag stat (SS-03)
        - [ ] Weapon_Might = the Might value on the equipped weapon (tome/magic weapon)
        - [ ] Magic_Triangle_Damage_Bonus = +1 if attacker has magic triangle advantage; −1 if disadvantage; 0 if neutral (WS-02, Layer 05)
        - [ ] Defender_Resistance = defender's Res stat (SS-08)
        - [ ] Note: Terrain Defense Bonus does NOT reduce magical damage. Only physical damage is reduced by terrain defence.
    - [ ] Magical damage minimum is 0.
    - [ ] Magical damage applies when the attacker is using a weapon tagged as magical (weapon_damage_type = MAGICAL in the weapon record).
- [ ] Critical Hit damage multiplier (interaction with CC-07):
    - [ ] If a critical hit triggers, final damage = base damage (from either formula above) × 3.
    - [ ] The ×3 multiplier is applied after all other modifiers. The base damage is computed first (including all bonuses and terrain), then multiplied.
    - [ ] Critical hit damage is not capped. A crit can deal more than a unit's max HP.
    - [ ] Critical hit damage minimum is still 0 (if base damage was 0 before the multiplier, the result is 0 × 3 = 0).
- [ ] Effective weapon damage bonus (WS-06, Layer 05 — interaction note):

    - [ ] If the attacking weapon has an effectiveness tag and the defending unit matches the effectiveness condition (e.g., bow vs. flying unit), the Weapon_Might used in the formula is tripled: effective_might = Weapon_Might × 3. The standard damage formula is then applied with this multiplied might value.
    - [ ] Effective weapon bonus and critical hit multiplier do not stack. If both trigger simultaneously (which is mechanically possible), critical hit takes precedence: base damage (calculated with effective might) × 3.
- [ ] Damage application:
    - [ ] Final computed damage is subtracted from the target unit's current HP.
    - [ ] Current HP minimum after damage is 0. HP cannot go below 0.
    - [ ] If current HP reaches 0, the unit is dead. Death handling is passed to Layer 07 (UM-01).

CC-08 — Double Attack (Speed Threshold) | Cat 1 | Core Combat

- [ ] Double attack eligibility:
    - [ ] A unit is eligible to attack twice in a single combat exchange if their effective Attack Speed (AS) is at least 4 points higher than their opponent's effective AS.
    - [ ] Double attack threshold: AS_attacker − AS_defender ≥ 4.
    - [ ] This check is performed for both combatants independently. Both the attacker and the defender can potentially double the other in the same combat exchange.
    - [ ] If the AS difference is exactly 4: double attack triggers.
    - [ ] If the AS difference is 3 or less: no double attack.
- [ ] Double attack application within combat round resolution:
    - [ ] See CC-09 for the full round resolution sequence. Double attack adds a second hit for the eligible unit in their designated position in the sequence.
    - [ ] A double attack hit uses the same weapon, same hit/damage formulas, and is subject to the same hit/miss/crit resolution as the first hit. It is a separate, independent RNG resolution.

    - [ ] A double attack hit can independently crit or miss relative to the first hit.
- [ ] Brave weapons (WS-08, Layer 05) override — interaction note:
    - [ ] Brave/Sarvasva weapons attack twice before the opponent counterattacks, regardless of AS difference. This is a weapon-triggered double, not an AS-triggered double. The AS double check is separate and can apply in addition. See WS-08 for full Brave weapon combat resolution.

CC-09 — Combat Round Resolution Order | Cat 1 | Core Combat

- [ ] Combat initiation:
    - [ ] Combat begins when a player unit attacks an enemy unit (or when an enemy AI decides to attack a player unit during Enemy Phase).
    - [ ] Before combat animations play, the combat forecast is computed and displayed (UI-03, Layer 20 — the forecast UI is a dependency stub here; combat resolution must function correctly even if the UI is not yet built).
    - [ ] The unit initiating the attack is the Attacker. The other unit is the Defender.
- [ ] Standard combat round resolution sequence:
    - [ ] Step 1 (Attacker's first hit): Attacker performs one attack against Defender. Hit/miss and damage are resolved per CC-04, CC-05, CC-06. If hit: damage is applied. If miss: no damage. Crit check: if the hit connects, check for critical hit (CC-07). If the Defender's HP reaches 0: combat ends immediately, no further steps execute.
    - [ ] Step 2 (Defender's counterattack): If the Defender is alive and has a weapon equipped with a range that includes the Attacker's current tile: Defender performs one attack against Attacker. Same resolution as Step 1. If the Attacker's HP reaches 0: combat ends immediately.
    - [ ] Step 3 (Attacker's double attack): If the Attacker qualifies for a double attack (CC-08) AND both units are still alive: Attacker performs a second attack. Same resolution as Step 1. If the Defender's HP reaches 0: combat ends.

    - [ ] Step 4 (Defender's double attack): If the Defender qualifies for a double attack AND both units are still alive: Defender performs a second attack. Same resolution as Step 1.
    - [ ] Combat ends after all qualifying steps are resolved, or at any point where either unit's HP reaches 0.
- [ ] Counterattack eligibility:
    - [ ] The Defender can counterattack only if their equipped weapon's range includes the distance between their tile and the Attacker's tile.
    - [ ] If the Defender has no weapon equipped, or their equipped weapon cannot reach the Attacker's range, no counterattack occurs (Steps 2 and 4 are skipped for the Defender).
    - [ ] Range is measured in grid tiles (Manhattan distance).
- [ ] Brave weapon interaction (WS-08):
    - [ ] If the Attacker is using a Brave/Sarvasva weapon, the sequence is modified: Step 1a (Attacker first hit) → Step 1b (Attacker second hit from Brave) → Step 2 (Defender counterattack) → Step 3 if AS double applies (Attacker third hit) → Step 4 if AS double applies for Defender.
    - [ ] The Defender cannot double before the Attacker has had both Brave hits, even if the Defender has higher AS.
- [ ] Staff-only units:
    - [ ] If the Defender has a staff equipped and no other weapon, they cannot counterattack. Staves are not offensive weapons in this context. Steps 2 and 4 are skipped.
    - [ ] Initiating combat with a staff is not possible. Staves are used via the "Staff" action in the action menu, not the "Attack" action.
- [ ] EXP award on combat resolution:
    - [ ] EXP is awarded to the player-controlled unit(s) involved in combat at the conclusion of the combat round.
    - [ ] EXP calculation is defined in Layer 07 (UM-03). At this layer, the combat system must fire a combat_resolved event passing: attacker_id, defender_id, attacker_survived (bool), defender_survived (bool), defender_killed (bool),

damage_dealt_by_attacker, and whether the attacker_hit on any of their strikes.
    - [ ] EXP is not awarded if a unit attacks a unit on the same faction (friendly fire scenarios, which can occur through Berserk status effects — Layer 06 onwards).

EDGE CASES

- A unit with 0 Might weapon (such as a staff accidentally equipped to a mage — prevented by weapon whitelist rules at equip, but edge case if data entry error exists): damage formula produces 0 + stats + bonuses. Not 0 total damage; the Str/Mag contribution still applies. Minimum damage is still 0 after defense subtraction.
- Two units with identical AS: no double attack for either. AS difference = 0, which is less than 4.
- Defender has no weapon equipped at all (unarmed): no counterattack at any step. No error should fire. Steps 2 and 4 simply do not execute.
- Attacker's double attack would be their third hit (after Brave weapon's two hits): this is valid. The Attacker delivers a Brave hit 1, Brave hit 2, then if AS double applies, a third hit. This can result in up to 4 total player hits (Brave ×2, AS double ×1, Defender AS double ×1 is on the defender side).

CC-10 — True Hit System → Trust-Worthy RNG | Cat 2 | Core Combat

DESIGN INTENT
This mechanic serves ES3 ("Feel like the conscience of a kingdom, carrying the weight of every life your war touches") and the structural requirement that deaths feel earned and grievable, not arbitrary. The FE GBA True Hit system is a double-RNG formula that causes displayed hit percentages to behave more fairly than a single random roll — high-percentage hits almost never miss, low-percentage hits almost never land. FE GBA never tells the player this exists. This version retains the identical double-RNG formula but changes the public posture of the system: the fairness is acknowledged in-world, and if a character dies to a statistically unlikely hit, the game says so. The goal is that a player who loses a

character to a low-probability hit should feel grief, not rage at the system.

- [ ] True Hit calculation — the double-RNG formula:
  - [ ] For every hit resolution, generate two independent random integers (rand1 and rand2), each uniformly distributed in the range 0–99 (inclusive).
  - [ ] True Hit value = ((rand1 + rand2) / 2) rounded to the nearest integer. This is the standard FE True Hit formula.
    - [ ] Equivalently: True_Hit = (rand1 + rand2 + 1) / 2 using integer arithmetic (rounded up on 0.5).
  - [ ] A hit occurs if the attacker's displayed hit percentage is greater than True_Hit.
    - [ ] Displayed hit percentage = max(0, min(100, Attacker_Hit − Defender_Avoid)).
    - [ ] If Attacker_Hit − Defender_Avoid ≥ 100: hit always connects (100% displayed and True Hit cannot exceed 99).
    - [ ] If Attacker_Hit − Defender_Avoid ≤ 0: attack always misses.
  - [ ] Critical hit resolution uses a separate single RNG roll (not the True Hit double formula). Critical uses a single random integer in 0–99; crit triggers if crit_rate > that roll.
  - [ ] The True Hit formula is used for all offensive hit checks in the game — player attacks, enemy attacks, allied NPC attacks. There is no exception.
- [ ] Displayed percentage accuracy:
  - [ ] The percentage displayed on the combat forecast UI (UI-03, Layer 20) must use the True Hit output range as its basis, not the raw Attacker_Hit − Defender_Avoid value.
  - [ ] Specifically: the displayed percentage must accurately reflect the probability that True_Hit < displayed_value, not the probability that a single RNG roll < displayed_value.
  - [ ] The displayed value is computed from the raw hit difference (Attacker_Hit − Defender_Avoid) but presented as the True Hit expected outcome:
    - [ ] For raw hit difference H where 0 ≤ H ≤ 100, displayed True Hit % = H − (H × (100 − H)) / 100 rounded to nearest integer. (This is the standard FE True Hit approximation formula.)

    - [ ] Players who understand probability will observe the game hits more often than a single-roll system would. This is intentional. The system is not hidden — it behaves the way the lore describes.
- [ ] Post-death honesty — the unlikely death acknowledgment:
    - [ ] When a named unit (player faction or named allied NPC) is killed by an attack, check: was the True Hit value used for that killing blow less than 30?
        - [ ] To check this: at the moment of hit resolution, the True_Hit value computed by the double-RNG formula is compared to 30. If True_Hit < 30 AND the attack connected AND the target unit is now dead, set unlikely_death_flag = true on the death event.
    - [ ] When unlikely_death_flag = true, an additional line is delivered in the post-battle sequence (after the War's Ledger, Layer 07, Cluster A):
        - [ ] A single authored text line appears on screen as if written by a campaign scribe — styled in the same visual register as the War's Ledger (not a popup, not a system message): "The odds were against it. The outcome was not."
        - [ ] This line is the same for every unlikely death. It is not character-specific. It is a statement about the nature of probability and fate.
    - [ ] The line does not appear during combat. It does not interrupt the death animation. It surfaces only in the post-battle sequence.
    - [ ] If multiple named units die in the same chapter with unlikely_death_flag, the line appears once per eligible death, appended to the relevant War's Ledger entry for each unit.
- [ ] Lore entry — "The Mathematics of Fate":
    - [ ] After the player's first battle (regardless of outcome), a lore entry titled "The Mathematics of Fate" is added to the in-game lore compendium.
    - [ ] Unlock trigger: chapter_1_battle_complete flag set.
    - [ ] Content: The entry explains the double-RNG probability in in-world language (authored text about karma and

probability — content team writes this). It is not a tooltip or tutorial pop-up. It is found only in the compendium.
    - [ ] The player is not directed to this entry. It is discoverable, not announced.
    - [ ] The entry exists in the compendium regardless of whether the player ever reads it. It is persistent.

EDGE CASES

- The raw hit difference is exactly 0 (Attacker_Hit = Defender_Avoid). Displayed True Hit = 0%. Both rand1 and rand2 must be 0 simultaneously for a hit to connect (True_Hit would be 0, and 0 < 0 is false). The attack always misses. Consistent with FE GBA behaviour.
- The raw hit difference is exactly 100. Displayed True Hit = 100%. No RNG is drawn; the hit always connects. This is a special-cased short-circuit: if Attacker_Hit − Defender_Avoid ≥ 100, skip RNG, mark as hit.
- A generic unnamed enemy unit kills a player unit with an unlikely hit. Does unlikely_death_flag apply? Only to named units. Generic enemy deaths are not tracked by the death registry (Cluster A). The unlikely_death acknowledgment applies only to named units on the player or named allied factions.
- The unlikely_death line would appear on the same post-battle screen as the War's Ledger. The line is appended to the dead unit's War's Ledger entry (Left Column, under their name), not displayed as a separate screen element.

CC-07 — Critical Hit (3× Damage) → Moral-Weight Reframe | Cat 2 | Core Combat

DESIGN INTENT
This mechanic serves ES3 ("Feel like the conscience of a kingdom, carrying the weight of every life your war touches"). In FE GBA, a critical hit is spectacle — a visual flourish, a musical sting, a moment of power. The math is unchanged here (3× damage). What changes is the audio-visual-narrative layer: crits are tiered by the moral weight of who is being struck. A crit against a corrupted Asura feels different from a crit against a human enemy soldier, which feels different again from a crit that kills someone who had a name, a story, and a

chance to have been spared. The moral weight does not change the tactical outcome. It changes how the game says "this happened."

- [ ] Critical Hit Rate calculation:
  - [ ] Crit Rate (attacker) = (Attacker_Skill / 2) + Weapon_Crit_Bonus + Class_Crit_Bonus (if applicable)
    - [ ] Where: Weapon_Crit_Bonus = the Crit stat on the equipped weapon; Class_Crit_Bonus = any crit bonus granted by the attacker's class (e.g., Swordmaster equivalent, Layer 09 — 0 at this layer if those classes aren't yet built).
    - [ ] Crit Rate minimum is 0.
  - [ ] Crit Evasion (defender) = Defender_Niyati stat (SS-06). This is the stat formerly called Luck in FE GBA.
  - [ ] Displayed Crit Rate = max(0, Crit_Rate − Crit_Evasion). This is shown on the combat forecast.
  - [ ] Crit resolution: generate one random integer 0–99 (single RNG, not True Hit double-roll). If this value < displayed_crit_rate, critical hit triggers.
  - [ ] A critical hit can only trigger if the attack first hits (True Hit roll succeeds). If the attack misses, crit is irrelevant.
- [ ] Moral weight tier system:
  - [ ] Every unit in the game has a moral_weight_tier field in their unit record. This field is set at unit spawn time and does not change during battle.
  - [ ] Valid tier values: ASURA, SOLDIER, NAMED.
  - [ ] ASURA: Non-human enemies — corrupted entities, Asura-class monsters, supernatural opponents. Any unit with unit type tag MONSTER defaults to ASURA unless overridden.
  - [ ] SOLDIER: Human enemy units without a name, backstory, or plot significance. Generic infantry, cavalry, archers, mages on the enemy side. Any human-type enemy unit without a name defaults to SOLDIER.
  - [ ] NAMED: Any unit with a name, a portrait, a story introduction (pre-chapter scene, support conversation eligibility, or recruitment possibility). Named enemy commanders introduced in pre-battle scenes are NAMED. Recruitable enemy units are NAMED.

    - [ ] Player-faction units and allied NPC units cannot be critted in a way that triggers the moral_weight_tier system (this system applies to crits performed by the player against enemy-faction targets). For incoming crits against player units, see the counterattack and pain portrait systems in SS-01 and the combat round resolution.
- [ ] Critical hit animation and audio tiers:
  - [ ] ASURA tier crit:
    - [ ] Hitstop (combat animation freeze frame at moment of impact): 10 frames (approximately 0.17 seconds at 60fps).
    - [ ] Flash effect: standard crit flash — high-impact visual, stylised, appropriate for defeating a supernatural threat.
    - [ ] Audio: standard crit sound — decisive, forceful impact. Victory sting plays normally.
    - [ ] No portrait display, no name display.
    - [ ] No special post-battle log entry beyond standard combat data.
  - [ ] SOLDIER tier crit:
    - [ ] Hitstop: 20 frames (approximately 0.33 seconds at 60fps).
    - [ ] Flash effect: the crit flash is present but the colour temperature shifts — warmer, less stylised than ASURA.
    - [ ] Audio: the victory sting is absent. Replaced with a single low resonant tone (tabla or similar — one note, brief, unaccompanied). No fanfare.
    - [ ] Portrait display: if the enemy unit has a portrait asset, display it on screen for 1.5–2 seconds before the death animation completes. If no portrait asset exists (generic soldier), display a silhouette of their unit class sprite instead.
    - [ ] No text, no number pop accompanying the portrait display.
    - [ ] Post-battle: if the enemy unit has a name (even a generic name from a name pool), their name is passed to the War's Ledger death registry as a

SOLDIER_NAMED entry. If fully anonymous, they increment the anonymous soldier count.
    - [ ] NAMED tier crit:
        - [ ] Hitstop: 60 frames (exactly 1 second at 60fps).
        - [ ] Flash effect: the crit flash is present but significantly reduced in intensity — a sharp cut rather than an explosion of light.
        - [ ] Audio: complete silence at the moment of impact. The ambient battle music pauses or drops to near-zero volume for 1.5 seconds. No sting. No tone. The silence is the response.
        - [ ] Portrait display: the unit's full portrait is displayed at the centre of the screen for 3–4 seconds before the death animation completes. A single line of text appears below the portrait: the unit's name only. No stats. No class. Just the name.
        - [ ] The portrait remains visible as a still image while the death animation plays beneath it. The portrait fades out after 3–4 seconds.
        - [ ] Post-battle: the death is logged in the War Chronicle as a "fate-sealed" event. The War's Ledger Left Column entry for this unit includes the "fate-sealed" designation. If a recruiting character was eligible to recruit this unit, a post-battle reaction line is flagged for delivery (see Cluster A Death Ripple system, Layer 07).
        - [ ] If a player unit who had expressed interest in recruiting this enemy (via a pre-battle line or prior Vakya attempt) is alive after the battle, their next camp scene dialogue includes a line: authored per-pair, reflecting the specific loss of someone who might have been an ally.
- [ ] War Chronicle integration:
    - [ ] On any crit that kills a unit, the system passes the following data to the War Chronicle (Layer 18, HM-03): attacker_unit_id, defender_unit_id (or anonymous count), chapter_id, moral_weight_tier, was_fate_sealed (boolean, true only for NAMED tier).

    - [ ] This data is used by the War Chronicle rendering system. The combat engine itself only needs to fire this event — it does not format the Chronicle entry.
- [ ] Hitstop values summary (for implementation clarity):
    - [ ] ASURA: 10 frames.
    - [ ] SOLDIER: 20 frames.
    - [ ] NAMED: 60 frames.
    - [ ] Hitstop duration is a configurable parameter per tier so it can be adjusted in testing without code changes.
    - [ ] Hitstop applies to the combat animation system, not to gameplay time. The game state does not change during hitstop; only the animation is paused.

EDGE CASES

- A crit triggers but does not kill the unit (unit survives at low HP). Tier-based presentation still applies (hitstop, audio, portrait display). The NAMED tier portrait and name display still play. The fact that the unit survived does not soften the moment — the weight is in the act of striking, not only in the death.
- A unit is NAMED tier but has no portrait asset loaded (content pipeline failure). Fall back to a silhouette of their class sprite, same as SOLDIER tier fallback. Do not display a blank or broken asset.
- Multiple crits happen in the same combat round (attacker crits on first hit, double attack also crits). Each crit triggers its own tier presentation. Both fire independently. The sequence is: first crit → full tier presentation → animation continues → second crit → full tier presentation. No compression or skipping of the second crit's tier sequence.
- A player unit is hit by a critical strike from an enemy (crit against the player). The moral_weight_tier system does not apply to incoming crits. The pain portrait system (SS-01) applies if HP drops to CRITICAL. The counterattack sound design (Cluster B B-08 reference) applies. No tier-based hitstop on incoming crits.

Layer 05 — Weapon System

Dependency: Layer 04 (Core Combat Engine) must be complete before this layer begins. Weapon properties are inputs to the combat formulas built in Layer 04.

WS-07 — Weapon Tiers (Iron/Steel/Silver/Killer) | Cat 1 | Weapon System

- [ ] Weapon tier data structure:
  - [ ] Each weapon has a tier field: IRON, STEEL, SILVER, or KILLER (and their cultural equivalents in naming — see naming conventions below).
  - [ ] Weapon tiers represent the quality/material of the weapon and determine a baseline expected power level for weapons of that tier.
  - [ ] Tier is a label — it does not directly affect any formula. Formulas read individual weapon stats (Might, Hit, Crit, Weight, Durability, Range). The tier is an organisational and UI-facing field.
- [ ] Iron tier weapons (cultural reskin name: Loha weapons — "iron" in Sanskrit):
  - [ ] Characteristics: low Might, high Hit, low Crit, low Weight, high Durability.
  - [ ] Baseline stats for iron sword equivalent: Mt 5, Hit 90, Crit 0, Wt 5, Uses 45.
  - [ ] Iron tier weapons are the starting weapons for most units. They are widely available in shops.
- [ ] Steel tier weapons (cultural reskin name: Tamra/Vajra weapons — harder metals):
  - [ ] Characteristics: higher Might than Iron, lower Hit, low Crit, higher Weight, moderate Durability.
  - [ ] Baseline stats for steel sword equivalent: Mt 8, Hit 75, Crit 0, Wt 10, Uses 30.
- [ ] Silver tier weapons (cultural reskin name: Rajata weapons — silver/refined):
  - [ ] Characteristics: higher Might than Steel, good Hit, low Crit, moderate Weight, low Durability.

    - [ ] Baseline stats for silver sword equivalent: Mt 13, Hit 80, Crit 0, Wt 11, Uses 20.
    - [ ] Silver tier weapons require weapon rank B or higher to equip (see WS-03).
- [ ] Killer tier weapons (cultural reskin name: Nishada weapons — the deathbringer):
    - [ ] Characteristics: modest Might, modest Hit, high Crit, moderate Weight, low Durability.
    - [ ] Baseline stats for killer sword equivalent: Mt 9, Hit 75, Crit 30, Wt 9, Uses 20.
    - [ ] Killer tier weapons require weapon rank B or higher to equip (see WS-03).
- [ ] Naming conventions:
    - [ ] All weapon names in the game use Indian mythology-inspired names. "Iron Sword" becomes "Loha Khadga," "Steel Lance" becomes "Tamra Shula," etc. The tier label (IRON, STEEL, SILVER, KILLER) is an internal data field. The display name is always the culturally-framed name.
    - [ ] A naming convention table mapping each weapon type × tier combination to its display name is a content asset, not a code asset. The code reads the display_name field from the weapon record.

WS-05 — Weapon Weight and Con Interaction | Cat 1 | Weapon System

- [ ] Weapon weight field:
    - [ ] Every weapon record has a weight (integer, minimum 1) field.
    - [ ] Weight interacts with the wielder's Con stat (SS-09) to compute the AS penalty (defined in SS-05 and SS-09):
        - [ ] AS penalty = max(0, weapon_weight − wielder_Con).
    - [ ] Weight is a fixed property of the weapon. It does not change.
- [ ] Weapon weight display:
    - [ ] Weapon weight is displayed on the weapon info panel in the inventory screen (UI-06, Layer 20) and on the combat forecast (UI-03, Layer 20).

    - [ ] On the combat forecast, if the weapon weight exceeds the wielder's Con (i.e., an AS penalty is active), the AS value is displayed with a visual indicator that it is reduced from the unit's base Speed (e.g., a downward arrow or a different colour on the AS value). The exact visual treatment is a UI Layer 20 decision; the data must be provided by the combat engine.

WS-06 — Effective Weapons (3× Damage vs Unit Types) | Cat 1 | Weapon System

- [ ] Effectiveness tag system:
    - [ ] Weapons can have an effectiveness_tag field listing unit type tags against which they deal triple Might.
    - [ ] Valid effectiveness targets: FLYING, ARMOURED, MONSTER, CAVALRY. (Additional tags may be added in content passes; the system must support any unit type tag as a potential effectiveness target.)
    - [ ] Effectiveness check: at combat initiation, if the attacking weapon has an effectiveness_tag that matches the defending unit's class type tag (UC-01), the weapon is "effective" against that unit.
    - [ ] When effective: the weapon's Might value in the damage formula is tripled: effective_might = weapon_Might × 3. The damage formula (CC-06) uses this tripled Might value.
    - [ ] Effectiveness is checked once at the start of combat resolution. It applies to all hits in the combat exchange (both Brave hits, both double attack hits, etc.).
- [ ] Effectiveness visual indicator:
    - [ ] When an effective weapon is used in an attack, the combat forecast UI displays an indicator that this weapon is effective against the target. The combat animation uses a distinct hit effect (a different impact visual — visually distinct from a normal hit, not as dramatic as a crit).
    - [ ] The indicator is shown on the forecast before combat, so the player knows in advance.
- [ ] Standard effective weapon types:
    - [ ] Bows (and bow equivalents — Dhanush class weapons): effective against FLYING units. Standard FE rule.

    - [ ] Armour-crushing weapons (specific weapon types tagged by content team): effective against ARMOURED units.
    - [ ] Light magic (Jyoti tomes): effective against MONSTER units (dark/corrupted entities).
    - [ ] Additional effective pairings are defined in the weapon record's effectiveness_tag and are content team decisions.

WS-09 — Ranged Tradeoff (1-2 Range Weapons) | Cat 1 | Weapon System

- [ ] Weapon range field:
    - [ ] Every weapon record has a range field. Range is expressed as a list of valid attack distances in tiles (e.g., [1] for melee-only, [2] for ranged-only, [1][2] for both).
    - [ ] Standard FE GBA range rules:
        - [ ] Melee weapons (range [1]): can only attack targets on adjacent tiles. Cannot be used to attack targets 2+ tiles away.
        - [ ] Ranged weapons (range [2]): can only attack targets at exactly 2 tiles distance. Cannot be used to attack adjacent (range 1) targets.
        - [ ] 1-2 range weapons (range [1][2]): can attack at either distance.
    - [ ] Range is checked at the time of attack target selection. Targets outside the weapon's valid range are not selectable.
- [ ] 1-2 range weapons — the tradeoff:
    - [ ] In FE GBA, 1-2 range weapons have lower Might or other reduced stats compared to single-range weapons of the same tier. This is the standard tradeoff: flexibility at the cost of raw power.
    - [ ] This game maintains this tradeoff. 1-2 range weapons of the same tier as single-range weapons have: −2 to −4 Might and/or reduced Hit compared to the single-range equivalent. Exact values are a balance pass decision.
    - [ ] 1-2 range weapons of the same tier exist for: swords with thrown-range variants (Tomara-type), lances with thrown range, and all magic tomes (all tomes have range [1][2] — magic can always be cast at melee or range).
- [ ] Ranged attack and counterattack range:

    - [ ] When a player unit attacks with a ranged weapon from 2 tiles, the enemy can only counter if the enemy's equipped weapon also has range [2] or [1][2].
    - [ ] An enemy with a melee-only weapon (range [1]) cannot counter a ranged attack from 2 tiles. No counterattack occurs in Step 2 of CC-09.
    - [ ] This is the core tactical tradeoff of ranged weapons: they prevent melee counterattacks.

WS-02 — Magic Triangle (Anima/Dark/Light) | Cat 1 | Weapon System

- [ ] Magic triangle definition:
    - [ ] Three schools of magic form a triangle with the same bonus/penalty structure as the physical weapon triangle:
        - [ ] Anima (elemental magic — wind, fire, thunder) → advantage over Dark
        - [ ] Dark (dark arts, Asura magic, entropy magic) → advantage over Light
        - [ ] Light (sacred magic, divine magic, purifying magic) → advantage over Anima
    - [ ] Magic triangle bonuses and penalties:
        - [ ] Advantage: +15 Hit, +1 Damage, +15 Avoid (same as physical weapon triangle advantage in WS-01).
        - [ ] Disadvantage: −15 Hit, −1 Damage, −15 Avoid.
        - [ ] Neutral: +0 to all.
    - [ ] Magic triangle applies when both combatants are using magic-type weapons (weapon_damage_type = MAGICAL). If one combatant uses physical and the other uses magic, neither triangle applies.
- [ ] Magic type tags on weapons:
    - [ ] Every magic weapon (tome, staff capable of damage — distinct from healing staves) has a magic_school field: ANIMA, DARK, or LIGHT.
    - [ ] Triangle advantage is computed by reading magic_school of attacker's weapon vs magic_school of defender's equipped weapon (if magical).
- [ ] Cultural vocabulary for magic schools:

    - [ ] ANIMA = Pancha Bhuta magic (the five elements: earth, water, fire, air, ether). Pancha Bhuta tomes are the Anima equivalents.
    - [ ] DARK = Tamasa magic (Tamas guna-aligned — dissolution, corruption, entropy). Tamasa tomes are the Dark equivalents.
    - [ ] LIGHT = Sattva magic (Sattva guna-aligned — purity, illumination, divine order). Sattva tomes are the Light equivalents.
    - [ ] These are display names and cultural framings. The underlying magic_school tag remains ANIMA, DARK, LIGHT for code purposes. All code reads the enum tag, not the display string.

WS-01 — Weapon Triangle | Cat 3 | Weapon System

Rolled Back to FE Original. Build the standard FE GBA weapon triangle behaviour.

- [ ] Physical weapon triangle definition:
    - [ ] Three physical weapon types form a circular advantage structure:
        - [ ] Sword (Kshara) → advantage over Axe (Stambha)
        - [ ] Axe (Stambha) → advantage over Lance (Vega)
        - [ ] Lance (Vega) → advantage over Sword (Kshara)
    - [ ] Triangle advantage bonuses (identical to FE GBA):
        - [ ] Advantage: +15 Hit, +1 Damage, +15 Avoid.
        - [ ] Disadvantage: −15 Hit, −1 Damage, −15 Avoid.
        - [ ] Neutral (same type vs. same type, or a weapon type not in the triangle such as Bow, Staff, or all tomes): +0 to all.
    - [ ] Triangle check occurs at the start of each combat initiation. The attacker's equipped weapon type is compared to the defender's equipped weapon type.
    - [ ] If the defender has no weapon, or a weapon not in the triangle (bow, tome, staff): no triangle applies.
    - [ ] Physical triangle applies only when the attacking weapon has weapon_damage_type = PHYSICAL.
- [ ] Weapon type tags:

    - [ ] Every physical weapon has a weapon_type field: SWORD (KSHARA), AXE (STAMBHA), LANCE (VEGA), BOW, or STAFF.
    - [ ] Cultural reskin: All UI-facing labels use the Sanskrit vocabulary (Kshara, Stambha, Vega) per Cluster B Mechanic B-09 (Shastra Mandala). The weapon_type enum codes internally use SWORD, AXE, LANCE for code clarity.
    - [ ] Codex entry "Shastra Mandala" unlocks after the first battle in which weapon triangle advantage plays a role (either player or enemy gains/suffers a triangle bonus). This is a content flag, not a UI interrupt.
- [ ] Triangle bonus application in formulas:
    - [ ] The triangle hit bonus (+15 or −15) is included in the Attacker Hit formula (CC-04) as the "Weapon Triangle Hit Bonus" term.
    - [ ] The triangle damage bonus (+1 or −1) is included in the Damage Calculation formula (CC-06) as the "Weapon_Triangle_Damage_Bonus" term.
    - [ ] The triangle avoid bonus (+15 or −15) is included in the Avoid formula (CC-05) — but it is the defender's triangle relationship that matters for their avoid, not the attacker's. Specifically: if the defender has triangle advantage over the attacker, the defender receives +15 Avoid.

WS-04 — Weapon Durability | Cat 3 | Weapon System

Rolled Back to FE Original. Build the standard FE GBA weapon durability behaviour.

- [ ] Durability field:
    - [ ] Every weapon record has a current_uses field (integer, minimum 0) and a max_uses field (integer, set at item creation).
    - [ ] current_uses starts equal to max_uses when a weapon is first obtained.
    - [ ] Each time a weapon is used in an offensive action (any attack, including misses), current_uses decreases by 1.
        - [ ] Exception: staff uses (healing, status staves) consume 1 use per successful activation, not per miss. Staves have no "miss" in the traditional sense.

        - [ ] For weapons with the Brave/Sarvasva mechanic (WS-08), each individual hit consumes 1 use. A full Brave round (2 hits) consumes 2 uses.
    - [ ] When current_uses reaches 0, the weapon breaks immediately after the hit that brought it to 0.
    - [ ] A broken weapon is removed from the unit's inventory permanently.
    - [ ] A weapon at current_uses = 1 is the last use. The "final use" happens on the next time it is used. There is no early warning before the final use beyond the current_uses counter being visible.
- [ ] Durability display:
    - [ ] current_uses is displayed on the weapon info panel (inventory screen, UI-06, Layer 20) and on the unit stat panel when the weapon is highlighted.
    - [ ] On the combat forecast (UI-03, Layer 20), the current_uses of the attacker's weapon is shown. If the weapon would break after this combat (current_uses = 1 and the attack will use 1), a visual indicator flags the impending break.
    - [ ] Broken weapons are not displayed in inventory — they are removed immediately on break.
- [ ] Indestructible weapons:
    - [ ] Certain weapons (specifically: legendary/character-locked weapons with the sacred_tier = LEGENDARY tag, Layer 07 and Cluster B B-13) may be flagged as indestructible.
    - [ ] If indestructible = true on a weapon, current_uses never decreases. The weapon cannot break.
    - [ ] Indestructible weapons are rare and plot-significant. The sacred weapon durability system from Cluster B B-11 applies to SACRED tier weapons, which are not indestructible but have the enhanced presentation layer (Layer 06+).

WS-03 — Weapon Ranks | Cat 3 | Weapon System

Rolled Back to FE Original. Build the standard FE GBA weapon rank behaviour.

- [ ] Weapon rank system:
  - [ ] Each unit has a weapon proficiency level for each weapon type they can equip, expressed as a rank: E, D, C, B, A, or S.
  - [ ] Rank is stored as a value per weapon type on the unit record. A unit only has rank entries for weapon types in their class's weapon type whitelist (UC-01). All other weapon types are inaccessible regardless of rank.
  - [ ] Starting rank for a newly recruited unit: determined by their unit record's starting_weapon_ranks field. Most units start at E or D rank. Experienced units may start at C or B.
  - [ ] Rank values in order: E (lowest) → D → C → B → A → S (highest).
- [ ] Rank-up through use:
  - [ ] Each time a unit uses a weapon in combat, their weapon rank experience (WEXP) for that weapon type increases by 1.
  - [ ] WEXP thresholds for rank-up (standard FE GBA values):
    - [ ] E → D: 1 WEXP
    - [ ] D → C: 40 WEXP
    - [ ] C → B: 71 WEXP
    - [ ] B → A: 121 WEXP
    - [ ] A → S: 201 WEXP
  - [ ] On rank-up, WEXP resets to 0 for the new rank. The unit's rank for that weapon type increases by one step.
- [ ] Weapon rank requirements on weapons:
  - [ ] Each weapon record has a min_rank field: E, D, C, B, A, or S.
  - [ ] A unit can equip a weapon only if their current rank in the weapon's type equals or exceeds the weapon's min_rank.
  - [ ] Weapons with no rank requirement use min_rank = E.
  - [ ] Attempting to equip a weapon above the unit's current rank results in the equip action being unavailable. The

weapon is displayed in inventory as locked/unavailable with the rank requirement shown.

- [ ] Rank display (cultural vocabulary — Cluster B B-10 reference):
  - [ ] On the unit stat screen (UI-02, Layer 20), weapon ranks are displayed using the Shishya Bond stage names rather than letter ranks:
    - [ ] E / D rank → displays as "Abhyasa" (practice)
    - [ ] C rank → displays as "Nishtha" (dedication)
    - [ ] B rank → displays as "Praveena" (proficiency)
    - [ ] A rank → displays as "Nipuna" (mastery)
    - [ ] S rank → displays as "Acharya" (one who becomes a teacher)
  - [ ] The letter rank (E/D/C/B/A/S) is used internally for all computation. The stage name is a display string only.
  - [ ] Rank-up acknowledgment: when a unit's weapon rank increases, a non-intrusive notification line appears in the character's voice (drawn from rank_up_acknowledgment[unit_id][weapon_type] pool — authored per-character). This fires once per rank-up. It is displayed as a small text overlay near the unit's portrait, 2.5 seconds, no pause. Generic units (unnamed enemies) do not receive rank-up acknowledgments.
- [ ] Weapon rank carry-through on promotion:
  - [ ] When a unit promotes to a new class (Layer 09), their weapon ranks are preserved exactly. Promotion does not reset weapon ranks.
  - [ ] If the promoted class grants access to a new weapon type the unit did not have before, the unit's rank in that new weapon type starts at E.

WS-08 — Brave Weapons | Cat 3 | Weapon System

Rolled Back to FE Original. Build the standard FE GBA Brave weapon behaviour. Note: the Sarvasva framing from Cluster B B-12 applies to the presentation layer and weapon naming conventions, but the mechanical rules are identical to FE GBA.

- [ ] Brave weapon mechanic:

    - [ ] A weapon with the brave_weapon_flag = true in its record causes the wielder to attack twice before the opponent counterattacks, in each combat exchange they initiate.
    - [ ] Modified combat resolution sequence for a unit using a Brave weapon (replacing CC-09 standard sequence):
        - [ ] Step 1a: Attacker's first hit (Brave hit 1). Resolved per CC-06, CC-04, CC-05, CC-07.
        - [ ] Step 1b: Attacker's second hit (Brave hit 2). Independent RNG. Independent crit check. Resolved identically to Step 1a. If the Defender is dead after Step 1a, Step 1b does not execute.
        - [ ] Step 2: Defender counterattack (if alive and in range). Standard resolution.
        - [ ] Step 3: If Attacker qualifies for AS double attack (CC-08), Attacker's third hit. Standard resolution.
        - [ ] Step 4: If Defender qualifies for AS double attack, Defender's second counter hit. Standard resolution.
    - [ ] The Brave weapon's two pre-counter hits each consume 1 durability use (WS-04). A full Brave attack round can consume up to 2–4 uses (depending on double attack eligibility).
    - [ ] Brave weapons are not limited to attack initiations: if a unit with a Brave weapon is the Defender, and they have counterattack range, they still only counterattack once (Steps 2 and 4 as normal). The Brave double-hit applies only when the unit is the Attacker initiating combat.
- [ ] Brave weapon acquisition:
    - [ ] Brave weapons are rare. In FE GBA, they are limited to specific high-tier items. This game follows the same rarity convention. Brave weapons are not sold in standard shops. They are found in specific chests, dropped by specific enemies, or rewards in specific story events.
    - [ ] Brave weapons use Silver-tier stat baselines with reduced Might compared to standard Silver weapons of the same type (tradeoff for the double-hit advantage).

WS-10 — Legendary / Character-Locked Weapons | Cat 3 | Weapon System

Rolled Back to FE Original for the mechanical lock system. The Cluster B B-13 presentation layer (covenant introduction scene, loss event, War Chronicle logging) is specified here as the authoritative source since it applies to weapon acquisition and loss events.

- [ ] Character-lock mechanic:
  - [ ] A weapon with character_locked = true in its record can only be equipped by a specific unit, identified by the weapon's owner_unit_id field.
  - [ ] If any unit other than the designated owner attempts to equip the weapon, the equip action is unavailable. The weapon can be held in inventory by other units (via Trade) but not equipped.
  - [ ] If the owning unit is dead (permadeath, Layer 07), the character-locked weapon remains in the dead unit's slot at the chapter end but is not accessible. After the chapter ends, it is logged in the War Chronicle as lost with the unit and is removed from the game.
- [ ] Legendary weapon properties:
  - [ ] All character-locked weapons are automatically tagged with sacred_tier = LEGENDARY (see WS-04 and the indestructible flag). They cannot break through normal durability loss.
  - [ ] Legendary weapons have stat profiles significantly above Silver tier: high Might, good Hit, modest Crit, and often special properties (effectiveness tags, magic triangle bypasses, or special animation triggers) defined per weapon in the weapon record.
  - [ ] Legendary weapons count as "S-rank equivalent" for equip purposes — only units who have reached S-rank (Acharya) in the relevant weapon type can equip them, unless the weapon's min_rank is overridden per specific story design.
- [ ] Covenant introduction scene (Cluster B B-13 implementation):
  - [ ] When a character-locked legendary weapon enters a unit's inventory for the first time (first acquisition,

regardless of method: story drop, chapter clear reward, promotion gift), fire the legendary_weapon_acquired event.
    - [ ] legendary_weapon_acquired event triggers a covenant_introduction_scene[unit_id][weapon_id]:
        - [ ] Scene type: portrait-with-background sequence (not a full map cutscene). Duration 30–45 seconds.
        - [ ] Content: the owning character speaks about the weapon — its history, who gave it to them (or how they came to carry it), and what they have sworn to do with it. All content is authored.
        - [ ] Scene is not skippable on first acquisition.
        - [ ] A lore entry for the weapon is added to the compendium at the same moment: the weapon's history, prior wielders, and origin. This entry is in the weapon record's lore_entry field (authored content).
    - [ ] After the first acquisition, the scene does not play again (flag: covenant_introduction_played[unit_id][weapon_id] = true).
- [ ] Weapon loss event (Cluster B B-13 implementation):
    - [ ] On the event legendary_weapon_depleted (which cannot occur for indestructible weapons unless specifically triggered by a story event that forces it) or legendary_weapon_lost (story-forced loss, unit death):
        - [ ] Set weapon_covenant_broken_flag[unit_id][weapon_id] = true.
        - [ ] At the start of the next chapter, trigger post_loss_dialogue[unit_id][weapon_id] (authored per weapon per character, 2–4 lines, portrait scene, skippable).
        - [ ] Filter the character's battle_dialogue_pool[] to exclude lines in the confident_tier category. Activate subdued_tier replacements for the rest of the campaign.
        - [ ] Check all allied units with support ≥ B-rank with the affected character. 50% chance to trigger weapon_loss_acknowledgment[observer_id][affected_id]. Fires max once per loss event per pair.
        - [ ] Log to the War Chronicle: pass weapon_id, weapon_name, bearer_name, chapter_id, loss_reason

with entry_type = WEAPON_LOST.

EDGE CASES (WS-10)

- The owning unit of a character-locked weapon is not on the current map (left in reserve). The weapon is in their inventory in the convoy. It cannot be traded to another unit for equipping. It can be traded to another unit's inventory slot but will remain unequippable until the owner is deployed.
- The owning unit is killed mid-battle. The weapon remains in their inventory for the duration of that battle (other units can Trade it to carry it, but not equip it). At chapter end, it is processed through the weapon_loss event system.
- A character-locked weapon is at 0 uses (impossible for indestructible, but consider edge case in testing): indestructible = true prevents this. If by data error it is not flagged indestructible and somehow reaches 0 uses, trigger the legendary_weapon_depleted event rather than silently removing it.